

MUHAMMAD IBRAHIM KHAN TAREEN

AI / ML ENGINEER • MLOPS • DEEP LEARNING • RAG

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TECHNICAL SKILLS

ML / AI

PyTorch, scikit-learn, NumPy, Pandas, RAG, LangChain

Languages & Tools

Python, C/C++, SQL, Jupyter

Web / Mobile

React Native, Node.js, LangGraph

Focus Areas

Enterprise Solutions, Mobile Products, Consumer Tech

EDUCATION

LUMS

BS Computer Science

2022 - 2026

Focus: Machine Learning, Generative AI, Deep Learning, Data Mining, MLOps

Roots International School

GCE A Levels

Mathematics (A), Computer Science (A), Physics (B), Further Mathematics (C)

COURSEWORK

Artificial Intelligence, Machine Learning, Deep Learning, Data Mining, Generative AI, Algorithms, Data Structures, Operating Systems, Computer Networks, Software Engineering, Digital Logic Design, Microcontrollers, Compiler Design, Circuits I

HIGHLIGHTS

- Strong foundation in ML evaluation, overfitting control, stratified splits, learning curves, and error analysis.
- Comfortable with end-to-end ML production workflows from raw preparation to robust pipeline scaling.

SUMMARY

AI/ML-focused Computer Science undergraduate at LUMS with hands-on experience in supervised learning, deep learning architectures, advanced RAG frameworks, and MLOps workflows. Proven capability in building production-ready pipelines, optimizing enterprise models, and handling end-to-end Machine Learning lifecycles with data-driven evaluation metrics.

EXPERIENCE

Systems Limited

Jul - Sep (8 weeks)

AI Products Intern

Lahore, Pakistan

- Built and evaluated supervised learning baselines for classification and regression models; conducted disciplined comparative analysis using accuracy, macro-F1, confusion matrices, and RMSE.
- Executed end-to-end data preparation pipelines, including data cleaning, robust preprocessing, and reusable feature engineering in Python (missing-value imputation, feature normalization, and encoding).
- Supported systematic experiment documentation and reproducible reporting protocols, establishing clean baseline parameters for upcoming architecture iterations.

PROJECTS

Crop Disease Classification (Final Year Project)

2025 - 2026

- Engineered an automated multi-crop disease classification pipeline leveraging traditional machine learning and advanced AI techniques to drive early agro-anomaly detection.
- Curated robust image datasets, implemented class balancing strategies, and established rigorous evaluation criteria via stratified train/validation/test splits.
- Investigated complementary architectures and confidence-based ensembling frameworks to achieve better real-world generalization and avoid validation overfitting.

MLOps & LLM Engineering Frameworks

Independent

- Designed fully custom AI agent workflows using LangChain and LangGraph, embedding LLMs, precise function calling, and structured orchestration pipelines.
- Developed highly reliable Retrieval-Augmented Generation (RAG) pipelines complete with optimal document chunking, semantic embeddings, vector database retrieval, and grounded response generation.
- Built reproducible orchestration workflows integrating experiment tracking tools and clean, modular interfaces for rapid system integration.

Deep Learning & Data Mining Portfolio

Academic

- Implemented state-of-the-art architectures including SRGAN, Vision Transformers (ViTs), standard Autoencoders, and Variational Autoencoders (VAEs).
- Deployed classical models and data mining tasks from scratch, encompassing Random Forests, Graph Convolutional Networks (GCNs), frequent pattern mining, and mini-RAG structures.

GITHUB PROJECTS

HealthSync

React Native

Architected a role-based healthcare application allowing synchronized doctor appointments, secure user flows, and real-time messaging across interfaces.

Multiplayer Battleship

Socket-based C++

Created a low-latency networked battleship game using standard socket connections with custom turn synchronization and resilient client-server message handling.